

By double negation law, $Q(a)$ is True. By defn. of conditional, $Q(a) \rightarrow \neg R(a)$ is False. From the defn. of universal quantifier, $\forall x (Q(x) \rightarrow \neg R(x))$ is False.

$\therefore d$ does not follow from a, b, c

By defn. of AND, $(\neg R(a) \vee S(a)) \wedge (\neg Q(a) \vee S(a))$ is True. By ^{1st} commutative law, $(S(a) \vee \neg R(a)) \wedge (S(a) \vee \neg Q(a))$ is True. By 1st Distributive Law,

$S(a) \vee (\neg R(a) \wedge \neg Q(a))$ is True. By 1st commutative law,

$\neg S(a) \vee \neg P(a)$ is True. Let, $S(a)$ be True. By defn. of negation, $\neg S(a)$ is False. By defn. of OR, $\neg P(a)$ is True. By defn. of conditional,

$[S(a) \Rightarrow \neg P(a)]$ is True. Let $S(a)$ be False. By defn. of negation,

$\neg S(a)$ is True. By defn. of OR, $\neg R(a) \wedge \neg Q(a)$ is True. By 2nd De Morgan's Law, $\neg(R(a) \vee Q(a))$ is True. By defn. of conditional,

$[\neg S(a) \Rightarrow \neg(R(a) \vee Q(a))]$ is True.

If something is not willing to waltz, then that thing is not an officer and also not one of my poultry. $\square$